The second-order asymptotic of the EKG sequence

P. Graham
https://orcid.org/0009-0007-8493-1505

October 24, 2026

Abstract

For the EKG sequence $a(n)$, let $b(n)$ replace values p and $3p$ by $2p$ for odd primes p. We prove $b(n) = n + n/(3\log n) + o(n/\log n)$ along all indices, giving the formula in Lagarias, Rains and Sloane's 2002 PDF abstract for $a(n)$ at almost all indices. Building on Haver's frontier reduction and sharp upper bound, the proof supplies the lower bound by an elementary sieve and transfers the frontier asymptotic to every smoothed position.

The proofs, the Lean formalization and the text were produced by AI models (from OpenAI and Anthropic), working under the author's direction.

1 The result

The EKG sequence starts $1, 2, 4, 6, 3, 9, 12, 8, 10, 5, 15, 18, 14, 7, 21$. Write $U_t = \{a(1), \ldots, a(t)\}$. Its defining greedy rule is

$$a(1) = 1, \quad a(2) = 2, \quad a(n) = \min\{m \geq 1 : m \notin U_{n-1},\ \gcd(m, a(n-1)) > 1\} \quad (n \geq 3). \tag{1}$$

We use the smoothing convention of LRS Figure 5 [9]:

$$b(n) = s(a(n)), \qquad s(m) = \begin{cases} 2p, & m = p \text{ or } m = 3p \text{ for an odd prime } p, \\ m, & \text{otherwise.} \end{cases} \tag{2}$$

All logarithms are natural; $\mathbb{N}$ denotes the natural numbers.

Theorem 1. *For the sequence defined by (1) and (2),*

$$b(n) = n + \frac{n}{3\log n} + o\left(\frac{n}{\log n}\right) \qquad (n \longrightarrow \infty).$$

Precisely, for every real $\varepsilon > 0$ there is an $N \in \mathbb{N}$, $N \geq 3$, such that for every $n \in \mathbb{N}$ with $n \geq N$,

$$\left| b(n) - n - \frac{n}{3\log n} \right| \leq \varepsilon \frac{n}{\log n}. \tag{3}$$

This is an eventual assertion along all indices of b, with one cutoff for each ε. The original sequence has the displayed formula only at almost all indices: the prime and three-times-prime branches remain different. Corollary 2 and its proof give the branch formulas and the density-zero exception set.

The question and prior work. Lagarias, Rains and Sloane's arXiv v1 PDF [9, abstract] says "We conjecture that almost all $a(n)$ satisfy the asymptotic formula" and displays $a(n) = n(1 + 1/(3\log n)) + o(n/\log n)$. Their opening credits the sequence as "discovered by Jonathan Ayres"; Section 4 says of the three lines, "This was also observed by Ayres". Their arXiv Conjecture 2 uses ratio asymptotics, a weaker assertion. Hofman and Pilipczuk's smoothed sequence a'_n replaces $p, 3p$ by $2p$ for every prime p, including 2. Their Theorem 4.17 [8] gives $a'_n = n + O(n/\log\log\log n)$. This convention differs from b at only finitely many indices. Haver's version 3 [7, Corollary 2(e)] states the counting criterion; his Theorem 1(d) and Corollary 12(d) give the sharp upper bound $E(t) - t \le (1/3 + o(1))t/\log t$, where $E(t)$ is the least unused positive even integer after t terms. Here the new work is the remaining lower bound and the transfer to all smoothed indices. The hard step is improving the displacement estimates from $O(x/\log x)$ to $o(x/\log x)$; multiplying the existing bounds leaves a fixed coefficient. The proof below supplies sparsity of the prime cofactors that those bounds alone do not detect.

2 Method and consequences

The argument proceeds from two uniform displacement estimates to exact counting, then to the sequence. Put $L(x) = x/\log x$ for $x > 1$. Propositions 6 and 7 show that used values other than $3p$ exceed $E(t)$ by at most $o(L(E(t)))$, and that every unused composite below $E(t)$ lies within $o(L(E(t)))$ of it. For a fixed $\delta > 0$ and a putative displacement $D \ge \delta x/\log x$, selected even multiples force distinct neighbors to have the form rq, where r is their common prime factor and q is prime in a common interval. Their positive density contradicts Lemma 4, a fixed-modulus sieve bound. The order of choosing constants makes the estimates uniform.

The exact identity (19) then has negligible error terms. The prime number theorem gives its coefficient $1 - 2(1/2) + 1/3 = 1/3$. An ordinary value is one unchanged by smoothing; a prime block is the pattern $2p, p, 3p$ for an odd prime p. At an ordinary position the pre-occurrence frontier handles a possible jump; at a prime block the three smoothed values coincide. Section G completes the proof, including the quantifiers of (3). All details follow in the appendix.

The consequences are stated and proved in Section H: the eventual sign of $a(n) - n$, its below-diagonal count and the sharper complementary count, finiteness on the three critical lines, finiteness of A155549 and A064420, and finiteness of cycles of each bounded length. They include the Ordowski and Jasinski conjectures quoted there. The three-branch formulas are Corollary 2. No completeness of the finite OEIS lists is asserted.

3 Checked scope and reproduction

The formal theorem is `ekg_sharp_all_index` in `lean/SolutionEndpoint.lean`. Its statement is exactly: for every real ε, if $0 < \varepsilon$, there exists a natural N with $3 \le N$ such that, for every natural n, $N \le n$ implies (3), with $b(n)$ equal to `EKG.CanonicalEndpoint.b n`. Natural numbers in that inequality are coerced to real numbers, and the logarithm is `Real.log`. This is the independently defined `EndpointStatement` in `lean/Statement.lean`; `lean/Check.lean` checks the theorem against that type. Thus the formal quantifiers are exactly those of Theorem 1, including one cutoff for all later indices.

In `lean/EKG/EndpointDefinitions.lean`, the formal $a(n)$ is `EKG.canonical.a (n - 1)`. The canonical sequence is the zero-based `actualA` in `lean/EKG/Canonical.lean`, starting at $1, 2$; its

`chooseNext` is proved in `lean/EKG/Basic.lean` to select the least positive unused value sharing a prime factor with the preceding term, equivalently having gcd greater than 1. This gives (1) for $n \geq 1$; truncated natural subtraction also sets $a(0) = 1$, outside the theorem's range $n \geq N \geq 3$. The formal b applies `smoothValue` to a: its two branches select odd primes and exactly three times odd primes, respectively. It is precisely (2), leaving 2 and 6 fixed.

The declaration has no additional sequence or analytic hypotheses. The imported estimates are proved dependencies, not assumptions to be supplied by the reader. `lean/Check.lean` guards its actual axiom list: `propext`, `Classical.choice`, and `Quot.sound` (propositional extensionality, classical choice, and quotient soundness).

The standard Lake project in `lean/` checks Theorem 1 from the literal greedy rule, including existence, leastness and the smoothing convention. Haver's estimates and the prime number theorem are imported proved theorems, recompiled with the proof; their provenance and licenses are in `THIRD_PARTY.md`. The build checks each proof against its independently stated type and allows only Lean's standard logical axioms. The unrelated admitted statements in the imported prime-number-theorem project are outside the checked proofs. The detailed source map and axiom output are in `checks/details.tex` and `lean/README.md`.

The Lean-checked consequences are the eventual sign, below-diagonal asymptotic, density-one conclusion, three-line finiteness, the two exceptional-set finiteness statements, and bounded-cycle finiteness. The three-branch corollary and the sharper complementary count are written deductions, not separate Lean theorems. The trusted base is Lean's kernel and its standard logical axioms; the written deductions are supplied in full. No human expert has reviewed this work.

Install elan, then from the public folder run:

```
cd lean
lake exe cache get
lake build
```

The toolchain and dependencies are pinned in `lean/lean-toolchain` and `lean/lake-manifest.json`. The optional numerical table and its generator in `data/` are not part of the proof. `lean/README.md` identifies the exact theorem roots, independent statement checks and finite controls.

Acknowledgment and license. The author thanks Tomer Haver for the estimates, counting identity, reduction and accompanying Lean sources. Copyright © 2026 the author named in this paper. Apache-2.0 (see the repository's LICENSE), except for the third-party material listed in `THIRD_PARTY.md`.

A Definitions, interfaces and the three branches

This section fixes the indexing and smoothing conventions and derives the three branches before the detailed proof of the main theorem. For comparison with (1), the Lean construction in `EKG/Canonical.lean` is transliterated below (`Nat` denotes the natural numbers). It is zero-based internally; `chooseNext` selects the least positive unused eligible value, with existence and minimality proved in `EKG/Basic.lean`.

```
noncomputable def actualA (n : Nat) : Nat :=
  if n = 0 then 1 else if n = 1 then 2 else
  chooseNext (fun i : Fin n => actualA i.val)
    (actualA (n-1))
termination_by n
decreasing_by all_goals omega
```

The one-based interface in `EKG/EndpointDefinitions.lean` uses
`a n := EKG.canonical.a (n - 1)`. Its value at zero is an unused total extension. Section 3 explains the checked existence, gcd eligibility and minimality statements, not additional assumptions on this construction.

In particular, $s(2) = 2$ and $s(6) = 6$. OEIS A256417 includes the prime 2 in both replacements [11]; it differs from b only at indices $2, 4$, where our values are $2, 6$ and its values are $4, 4$. Here is the corresponding Lean definition, transliterated from `EKG/EndpointDefinitions.lean`, using ASCII type names:

```
def smoothValue (m : Nat) : Nat :=
  if And m.Prime (Not (m = 2)) then 2 * m
  else if And (m / 3).Prime
      (And (Not (m / 3 = 2)) (m = 3 * (m / 3))) then
    2 * (m / 3)
  else m
noncomputable def b (n : Nat) : Nat := smoothValue (a n)
```

Here `And` and `Not` are Lean's logical constructors; the distributed source uses the usual symbolic notation. All logarithms are natural.

The Lean type, transliterated using `abs`, `And` and `Exists` for the usual logical notation and opening `EKG.CanonicalEndpoint`, is

```
theorem ekg_sharp_all_index :
  forall eps : Real, 0 < eps ->
    Exists (fun N : Nat => And (3 <= N)
      (forall n : Nat, N <= n ->
        abs ((b n : Real) - (n : Real) -
          (n : Real) / (3 * Real.log (n : Real))) <=
        eps * ((n : Real) / Real.log (n : Real))))
```

The theorem has no sequence-estimate hypotheses. The separate statement in `Statement.lean` and its type check in `Check.lean` fix exactly these quantifiers and casts.

Corollary 2 (The three branches)**.** *Along the indices of each indicated kind, as* $n \to \infty$*,*

$$\begin{aligned} a(n) &= n + \frac{n}{3\log n} + o(n/\log n), && a(n) \neq p, 3p \quad (p \text{ odd prime}), \\ p &= \frac{n}{2} + \frac{n}{6\log n} + o(n/\log n), && a(n) = p, \\ 3p &= \frac{3n}{2} + \frac{n}{2\log n} + o(n/\log n), && a(n) = 3p. \end{aligned}$$

Proof. In the three cases $a(n) = b(n)$, $a(n) = b(n)/2$, and $a(n) = 3b(n)/2$, respectively. Multiply Theorem 1 by the fixed factor. The finitely many cases involving 2 do not matter for any limit. □

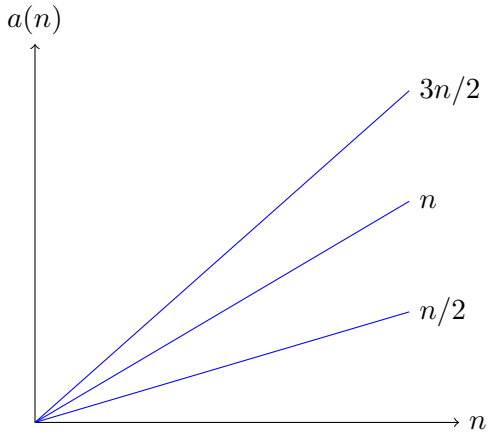


Figure 1: The three critical lines, shown schematically. This is not sequence data; Corollary 2 gives the positive second-order corrections, and Corollary 10 proves finiteness of intersections with these exact lines.

These are the three lines observed by Ayres and by Lagarias, Rains, and Sloane [9, Section 4]. For large k with $a(k) \in \{p, 3p\}$ we have $b(k) = 2p \leq 2k$, so at most $2\pi(N) + O(1) = o(N)$ indices $k \leq N$ are of these two kinds. Hence the first line gives the formula of the LRS abstract for almost all n: this is the additive second-order interpretation. It implies the literal Conjecture 2, whose ratio limits were already proved by Hofman and Pilipczuk [8, Theorem 4.17]. LRS Conjecture 3, $13n/28 \leq a(n) \leq 12n/7$ for every n, with equality only at $n = 28$ and $n = 7$, respectively, is not settled here: an eventual asymptotic does not settle the required finite range. It is proved, with computer assistance, in our companion paper [5]. Nor do we address the unproved distributional suggestions for higher-order fluctuations in LRS Section 4 or their unnumbered cycle questions, beyond the bounded cycle lengths of Section H.

Section H derives three consequences. For all large n, $a(n) > n$ exactly when $a(n)$ is composite, and $\#\{k \leq N : a(k) < k\} \sim N/(2\log N)$, so $\#\{k \leq N : a(k) > k\}/N \to 1$. This proves two 2009 conjectures of Thomas Ordowski recorded in OEIS A064413: that $a(n) > n$ almost everywhere when $a(n)$ is not prime, which our eventual form strengthens, and the density statement. The equations $a(k) = k$, $2a(k) = k$ and $2a(k) = 3k$ have finitely many solutions; this proves Artur Jasinski's 2008 conjectures that A152458 ($a(k) = k$) is finite and, on A152519, that the three lines carry finitely many points in total. And the permutation has finitely many cycles of each bounded length.

B Frontiers and established estimates

This section fixes the frontier notation and records the established estimates used in the proof.

Write $L(x) = x/\log x$ for $x > 1$, and let $U_t = \{a(1), \dots, a(t)\}$. Let $E(t)$ be the least positive even integer outside U_t. Call a value $3p$, for an odd prime p, a *tick*, and put

$$T(t) = \max\{a(i) : 1 \le i \le t,\ a(i) \text{ is not a tick}\}, \qquad K(t) = \#\{m \in U_t : 2 \mid m,\ m > E(t)\}.$$

The frontier $E(t)$ is nondecreasing and tends to infinity, by the permutation property. Here and below, $\pi(x)$ counts primes at most the real number x, and $\mathrm{spf}(h)$ is the least prime factor of h. We use the following previously established estimates.

Proposition 3 (Established sequence estimates)**.** *There are fixed constants $C \ge 1$ and $C_1 > 0$ such that, eventually, writing $e = E(t)$ and $k(t) = K(t) + 1$,*

$$|b(n) - n| \le CL(n), \tag{4}$$

$$k(t) \le K_0 \frac{\sqrt{e}}{\log e}, \qquad K_0 = 793, \tag{5}$$

$$T(t) - e \le C_1 L(e). \tag{6}$$

If $t \ge 9$ and a composite $15 \le h < e$ is outside U_t, then

$$e - h \le 2\,\mathrm{spf}(h)(13\pi(\sqrt{h}) + 724). \tag{7}$$

For all sufficiently large e,

$$F(e) := 13\pi(\sqrt{e}) + 724 \le B\frac{\sqrt{e}}{\log e}, \qquad B = 802. \tag{8}$$

Every composite below 15 is in U_{13}. For each odd prime p, its first multiple occurs at an index N_p, and

$$a(N_p) = 2p = E(N_p - 1), \quad a(N_p + 1) = p, \quad a(N_p + 2) = 3p. \tag{9}$$

Distinct introduction indices N_p are at least four apart.

These are results about the actual greedy sequence, not hypotheses added to Theorem 1. The prime pattern goes back to Hofman and Pilipczuk [8, Theorem 4.3]; we use its introduction formulation as proved in Haver's archive [7, Section 2, (F2)]. The early-even bound follows from Haver's bound $K(t) \le 12\pi(\sqrt{e}) + 720$ and elementary prime estimates. One may take $C_1 = 4 \cdot 793^2 + 2 \cdot 793$ in (6). The printed sources of (7) and (4) are Haver's Theorem 11 and Corollary 12(c), respectively. The constants in (5)–(8) are convenient weakenings proved in the accompanying sources, rather than optimized constants: $793 = 12 \cdot 6 + 721$ and $802 = 13 \cdot 6 + 724$ follow from the eventual bound $\pi(\sqrt{e}) \le 6\sqrt{e}/\log e$ and $\sqrt{e}/\log e \ge 1$, absorbing the additive constants. The prime estimate in (8) does not require the prime number theorem.

For the following comparison, write $e = E(t)$, let D denote a positive displacement, let r be the least prime factor of a predecessor, and let h denote a missing composite. The quantities $k(t), K_0, F(e)$ and spf have been defined above.

The coefficient has the counting heuristic of LRS Section 4 [9]:

$$n \approx x - \frac{x}{2\log x} + \frac{x}{6\log x} = x - \frac{x}{3\log x}.$$

Our new step upgrades two $O(x/\log x)$ displacement bounds to little-o. For the upper bound, the old greedy inequality is $D \le 2rk(t)$, with $r \le \sqrt{e+D}$ and $k(t) \le K_0\sqrt{e}/\log e$. Multiplying these estimates leaves a fixed coefficient of $e/\log e$; it does not force that coefficient to vanish. For the lower bound, $D \le 2\,\mathrm{spf}(h)F(e)$ and $\mathrm{spf}(h) \le \sqrt{e}$ have the same loss. Haver already identifies these rough-predecessor and rough-missing-composite obstructions in the remark after Corollary 12 [7]. The additional information is sparsity among the forced prime cofactors. Given a putative displacement $D \ge \delta x/\log x$, we instead select multiples $2rMj$ and force their distinct neighbors to be products rq_j with q_j prime. There are at least $D/(4rM) - O(1)$ such cofactors in an interval of length $O(x/(r\log x))$. For each fixed $\delta > 0$ their density is bounded below by a positive constant independent of x. It need not be independent of δ: M also depends on δ. The sieve bound permits an arbitrarily small fixed upper density, chosen after δ and M, which is exactly what the little-o conclusion needs.

The two estimates are an upper bound for the largest ordinary used value above the least unused even integer, and a lower bound on every unused composite below it. Sections D and E prove them, then exact counting and the prime number theorem give the result. We extract no explicit remainder rate. The nonquantitative step is the choice of a fixed sieve modulus for each δ, followed by unspecified eventual cutoffs in localization and the qualitative prime number theorem. This is not a claim of an intrinsically ineffective obstruction: the sieve constants are computable, but this proof does not supply a function $N(\varepsilon)$.

C Greedy bounds and interval prime capacity

This section supplies the greedy inequality and the elementary sieve tools for both displacement bounds.

If q is any prime divisor of $a(j-1)$, every positive multiple of q smaller than $a(j)$ was used before index j; otherwise the greedy choice would be smaller. Consequently, at any observation time t,

$$a(j) \le E(t) + 2qk(t) \qquad (3 \le j \le t+1). \tag{10}$$

Indeed, all multiples of $2q$ strictly between $E(t)$ and $a(j)$ are used even values above $E(t)$, and there are at least $(a(j) - E(t))/(2q) - 1$ of them. Their number is at most $K(t)$. Notice that (10) holds for every prime divisor q, at the same observation time, even if j is much earlier than t.

Euler's totient $\varphi(W)$ counts the reduced residue classes modulo W, and $\omega(W)$ counts its distinct prime divisors.

The following elementary fixed-modulus sieve estimate is included with its proof; compare the sieve of Eratosthenes in [1].

Lemma 4. *For every $\eta > 0$ there is a constant C_η such that every closed real interval of length $z \ge 0$ contains at most $\eta z + C_\eta$ primes.*

Proof. Choose a positive integer $Z \ge 2$ sufficiently large, and put $W = \prod_{p \le Z} p$. Unique factorization and geometric series give

$$\frac{W}{\varphi(W)} = \prod_{p \le Z} (1 - 1/p)^{-1} \ge \sum_{m=1}^{Z} \frac{1}{m} \ge \log(Z+1).$$

We may therefore choose Z with $\varphi(W)/W < \eta$. A residue class modulo W has at most $z/W + 1$ members in such an interval. Primes not dividing W lie in its $\varphi(W)$ reduced residue classes, and

at most $\omega(W)$ primes divide W. Thus $C_\eta = \varphi(W) + \omega(W)$ works. Both constants are fixed before the interval varies. □

Lemma 5 (Forced common factor)**.** *Let $x, \beta > 0$, let r be prime, and let $M \geq 1$ be an integer. Suppose every prime divisor of $2M$ is smaller than $\beta\sqrt{x}$, and $(\beta\sqrt{x})^3 > 2x$. For selected positive integers $j < \beta\sqrt{x}$, put $y_j = 2rMj$. Suppose the distinct integers d_j are composite, satisfy $d_j < 2x$, have every prime divisor at least $\beta\sqrt{x}$, and satisfy $\gcd(d_j, y_j) > 1$. Then $d_j = rq_j$ for distinct primes q_j. If all d_j belong to an interval I, the q_j belong to the interval I/r.*

Proof. Choose a prime ℓ dividing both d_j and $2rMj$. Every prime divisor of j is at most $j < \beta\sqrt{x}$; every prime divisor of $2M$ is smaller than that bound by assumption. Since all prime divisors of d_j are at least $\beta\sqrt{x}$, ℓ divides neither j nor $2M$. Euclid's lemma forces $\ell \mid r$, so $\ell = r$. A composite d_j has at least two prime factors counted with multiplicity. It cannot have three, since their product would be at least $(\beta\sqrt{x})^3 > 2x > d_j$. It therefore has exactly two, one of which is r; the other is a prime q_j, with $q_j = r$ allowed. Distinct d_j give distinct q_j by cancellation of the fixed positive integer r. Division by r gives the interval assertion. □

We also record how the old rate localizes adjacent ordinary values. If $y = a(j)$ is unchanged by smoothing and $e/2 < y < 2e$, then, uniformly as $e \to \infty$, the occurrence index j tends to infinity. To see this, choose any fixed index cutoff and exceed the maximum of the finitely many preceding values. Equation (4) then gives $j/2 \leq y \leq 3j/2$ and hence $e/3 < j < 4e$. If an adjacent value $d = a(j-1)$ or $a(j+1)$ is also unchanged by smoothing, the old rate at both indices gives

$$|d - y| \leq CL(j) + CL(j \pm 1) + 1 = O(L(e)). \tag{11}$$

The constant is uniform. This uses only the old big-O rate, not the sharp clock we will prove.

D The upper excursion

This section proves the uniform upper displacement bound for ordinary used values.

Proposition 6. *As $t \to \infty$, $\max\{T(t) - E(t), 0\} = o(L(E(t)))$.*

Proof. Fix $\delta > 0$. All constants in this proof are chosen before the final threshold on t. Write $e = E(t)$, $v = T(t)$, and suppose

$$D = v - e \geq \delta L(e). \tag{12}$$

The old span implies $v < 2e$ for sufficiently large e. Choose $i \leq t$ with $a(i) = v$, and let $c = a(i-1)$. The predecessor is not even: the unused even e would then be an eligible choice smaller than v. It is not 1, which appears only at the first index, and it is not an odd prime, since v would then be a tick. If it were a tick, (10) with $q = 3$ would give $D \leq 6k(t) = o(L(e))$, a contradiction. Thus c is an odd composite non-tick, so $c \leq v$. For $r = \mathrm{spf}(c)$, (10) and (5) imply

$$\alpha\sqrt{e} \leq r \leq \sqrt{v} < \sqrt{2e}, \qquad \alpha = \frac{\delta}{2B}. \tag{13}$$

Here $B \geq K_0$, so the displayed lower bound is a weakening of $r \geq \delta\sqrt{e}/(2K_0)$.

Set $\beta = \delta/(4K_0)$, and fix an integer $M \geq 2$ with $M > 1/(\alpha\beta)$. Consider

$$Y = \{2rMj : j \in \mathbb{N},\ j \geq 1,\ e + D/2 < 2rMj < v\}.$$

Every member of Y is a multiple of r smaller than v, so it occurred before i. Counting a lattice in an open interval gives

$$|Y| \geq \frac{D}{4rM} - 2. \tag{14}$$

For $y = a(i_y) \in Y$, put $d_y = a(i_y - 1)$. For every prime $q \mid d_y$, the common observation-time inequality (10) yields

$$q > \frac{D}{4k(t)} \geq \beta\sqrt{e}.$$

For large e this excludes 2 and 3. The predecessor is not 1, a prime, or a tick: in the prime case its successor would be odd, whereas y is even. Therefore d_y is an odd composite non-tick, and $d_y \leq v < 2e$. For large e, $(\beta\sqrt{e})^3 > 2e$.

Write $y = r(2Mj)$. Equation (13) gives

$$j < \frac{\sqrt{e}}{\alpha M} < \beta\sqrt{e}.$$

For large e, every prime factor of the fixed integer $2M$ is also smaller than $\beta\sqrt{e}$. Consecutive values d_y, y share a prime, and distinct selected values have distinct predecessor indices and hence distinct predecessors. Lemma 5 therefore gives

$$d_y = rq_y \quad \text{for distinct primes } q_y. \tag{15}$$

Both y and d_y are unchanged by smoothing. The finite-prefix argument preceding (11) applies uniformly to their indices. Since $e < y < v \leq e + C_1 L(e)$, it gives, for a fixed $A > 0$,

$$e - AL(e) \leq d_y \leq e + AL(e).$$

For example, $A = C_1 + 8C + 1$ suffices. Distinct y have distinct occurrence indices, distinct predecessor indices, and hence distinct d_y. Thus (15) injects Y into the primes in $[(e - AL(e))/r, (e + AL(e))/r]$. By Lemma 4,

$$\frac{D}{4rM} - 2 \leq \eta \frac{2AL(e)}{r} + C_\eta.$$

Choose $\eta = \delta/(32AM)$ after fixing M. It follows that

$$D \leq \frac{\delta}{4} L(e) + 4\sqrt{2}M(C_\eta + 2)\sqrt{e}.$$

Since $\sqrt{e} = o(L(e))$, the right side is less than $\delta L(e)$ for sufficiently large e, contradicting (12). The frontier tends to infinity, so one threshold on t works for every later prefix. This proves the proposition. □

E Missing composites

This section bounds the distance of every missing composite below the frontier.

Proposition 7. *For every $\delta > 0$ there is t_0 such that, for all $t \geq t_0$ and every composite $h < E(t)$ outside U_t, $E(t) - h < \delta L(E(t))$.*

Proof. Fix $\delta > 0$, put $e = E(t)$, and suppose such a missing h has $D = e - h \geq \delta L(e)$. Take $t \geq 13$. Then $h \geq 15$, and h is odd because every positive even below e is used. Put $s = \mathrm{spf}(h)$. Equations (7) and (8) imply

$$D \leq 2sF(e) \leq 2BL(e), \qquad \alpha\sqrt{e} \leq s < \sqrt{e}, \quad \alpha = \frac{\delta}{2B}. \tag{16}$$

Set $\beta = \delta/(6B)$ and fix $M \geq 2$ with $M > 1/(2\alpha\beta)$. Select the even multiples $y = 2sMj$ in $(e - D/2, e)$. They all belong to U_t. Discard $a(t)$ if it belongs to this set, losing at most one value in total, and call the resulting set Y. Then

$$|Y| \geq \frac{D}{4sM} - 3. \tag{17}$$

Each $y = a(i_y)$ has a successor $u_y = a(i_y + 1)$ with $i_y + 1 \leq t$. The value h is still unused and eligible after y, so $u_y \leq h$ by the greedy rule. Equality is impossible since h is missing at time t. Thus $u_y < h$.

By (16), $y > e - BL(e) > e/2$ for large e. The old rate and a finite-prefix maximum force all i_y beyond any fixed cutoff, uniformly in y; in particular they exceed the cutoff for Proposition 6 with parameter $\delta/6$. For $f_y = E(i_y) \leq e$, that proposition gives

$$f_y \geq y - \frac{\delta}{6}L(f_y) \geq y - \frac{\delta}{6}L(e).$$

We used that L is increasing for arguments at least 3. Consequently,

$$f_y - u_y > \frac{D}{2} - \frac{\delta}{6}L(e) \geq \frac{\delta}{3}L(e). \tag{18}$$

These are post-occurrence frontiers. Their divergence follows from the uniform divergence of the indices i_y.

The successor $u = u_y$ is unused at time i_y and below f_y, so it is not even. It is not an odd prime p: its predecessor would be $2p$, whereas $y/2 = sMj$ is composite. It is not a tick, whose predecessor is odd, and it is not 1. It is therefore an ordinary odd composite. Both y and u are unchanged by smoothing. Their occurrence indices satisfy $e/4 < i_y < 2e$ for large e, by the old rate. Applying that rate at $i_y, i_y + 1$ gives

$$|u - y| \leq (6C + 1)L(e), \qquad e - A_B L(e) < u < e, \quad A_B = B + 6C + 1.$$

In particular $u \geq 15$ for sufficiently large e, before we apply the missing-composite bound to it. Combining (7), now at time i_y, with (18) yields

$$\frac{\delta}{3}L(e) < f_y - u \leq 2\,\mathrm{spf}(u)F(e), \qquad \mathrm{spf}(u) > \beta\sqrt{e}.$$

Every prime factor of u is large, while

$$j < \frac{\sqrt{e}}{2\alpha M} < \beta\sqrt{e},$$

and every prime factor of the fixed integer $2M$ is smaller than $\beta\sqrt{e}$ eventually. Since $u < e < (\beta\sqrt{e})^3$ eventually, Lemma 5 gives $u = sq_y$ with q_y prime. This does not assert that $s = \mathrm{spf}(u)$.

Distinct selected y have distinct successor indices and values, so the q_y are distinct. They lie in $((e - A_B L(e))/s, e/s)$, an interval of length $A_B L(e)/s$. By (17) and Lemma 4,

$$\frac{D}{4sM} - 3 \le \eta \frac{A_B L(e)}{s} + C_\eta.$$

Fix $\eta = \delta/(16MA_B)$. Since $s < \sqrt{e}$,

$$D \le \frac{\delta}{4} L(e) + 4M(C_\eta + 3)\sqrt{e} < \delta L(e)$$

for sufficiently large e, a contradiction. The choices have the order δ, the upper-excursion cutoff, M, η, C_η, and finally the frontier and index cutoffs. All are uniform in the missing composite h and the selected y. □

F Exact counting and the frontier clock

We give the boundary terms explicitly, since the main coefficient arises from a difference of prime counts. Fix $t \ge 9$ with $e = E(t) \ge 14$, and put

$$\begin{aligned}
H &= \#\{h < e : h \text{ composite},\ h \notin U_t\},\\
O &= \#\{w \in U_t : w > e,\ w \text{ odd and not a tick}\},\\
P_1 &= \#\{p \text{ prime} : e/2 \le p < e\},\\
P_2 &= \#\{p \text{ prime} : e/3 < p < e/2\},\\
\theta_1 &= \mathbf{1}_{\{\exists \text{ odd prime } p<e/2 \text{ with } N_p=t\}},\\
\theta_2 &= \#\{p \text{ prime} : e/3 < p < e/2,\ N_p \in \{t-1, t\}\}.
\end{aligned}$$

Let $\theta = \theta_1 + \theta_2$; the spacing of introduction indices gives $0 \le \theta \le 2$.

For $p < e/2$, $2p < e$ is already used, so (9) shows that p is missing precisely when $N_p = t$. For $e/2 \le p < e$, $2p$ has not been used, so p is missing. For the first assertion the prime 2 is already used and is excluded. For the second, a used $2p \ge e$ would have entered at a frontier $2p = E(N_p - 1) \le e$, forcing the impossible used value e. There are therefore $H + P_1 + \theta_1$ missing positive integers below e. The used ticks above e are exactly those indexed by P_2, except for the θ_2 unfinished blocks. No used prime lies above e. Counting the t used values below and above the unused value e gives

$$t = (e - 1) - H - P_1 - \theta_1 + K + O + P_2 - \theta_2.$$

Write $I = \mathbf{1}_{\{e/2 \text{ prime}\}}$. Since e is even and at least 14,

$$P_1 = \pi(e) - \pi(e/2) + I, \qquad P_2 = \pi(e/2) - I - \pi(e/3).$$

We obtain the exact identity

$$\begin{aligned}
e - t = \pi(e) - 2\pi(e/2) + \pi(e/3)\\
+ H - K - O + 1 + \theta + 2I.
\end{aligned} \tag{19}$$

Haver proves this counting identity in Corollary 12(b). The old missing-composite estimate suffices to count holes. For $t \ge 13$, classify a missing composite by $s = \mathrm{spf}(h)$. By (7), it is an odd multiple

of the odd prime s in $[e - 2sF(e), e)$. There are at most $F(e) + 1$ such multiples, because their spacing is $2s$. Hence

$$H \le \pi(\sqrt{e})(13\pi(\sqrt{e}) + 725) = O(e/\log^2 e) = o(L(e)).$$

Equation (5) gives $K = o(L(e))$. Proposition 6 gives $O \le \max\{T(t) - e, 0\}/2 + 1 = o(L(e))$. The correction $1 + \theta + 2I$ is between 1 and 5.

The prime number theorem, in the form $\pi(x) \sim x/\log x$ [6, 3], implies for each fixed $c > 0$ that $\pi(ce)/L(e) \to c$. Thus

$$\frac{\pi(e) - 2\pi(e/2) + \pi(e/3)}{L(e)} \longrightarrow 1 - 2 \cdot \frac{1}{2} + \frac{1}{3} = \frac{1}{3}.$$

Equation (19) yields $e - t = L(e)/3 + o(L(e))$. Since $L(e) = o(e)$ and $e \to \infty$, this implies $e \sim t$, and consequently $L(e) \sim L(t)$. We have proved

$$E(t) = t + \frac{1}{3}L(t) + o(L(t)). \tag{20}$$

The bound $O = o(L(e))$ proves the remaining hypothesis of the equivalence stated in Haver's Corollary 2(e) and proved in Corollary 12(d); the counting criterion itself is not new here.

G Passage to the full sequence of indices

This section transfers the frontier asymptotic to all smoothed positions.

Proof of Theorem 1. First suppose $x = a(n)$ is ordinary, at a sufficiently large index, and use the pre-occurrence frontier $f = E(n - 1)$. If $x < f$, then x is odd, unused at time $n - 1$, and neither 1 nor a prime, so Proposition 7 gives $f - x = o(L(f))$, uniformly over these occurrences. If $x = f$, the discrepancy is zero. If $x > f$, the unused even f remains unused at time n, whence $E(n) = f$. Now $x \le T(n)$ and Proposition 6 gives $x - f = o(L(f))$. All three cases therefore give

$$a(n) = E(n - 1) + o(L(E(n - 1))).$$

By (20), $E(n - 1) \sim n$, $L(E(n - 1)) \sim L(n)$, and $E(n - 1) = n + L(n)/3 + o(L(n))$. This proves the desired formula at all ordinary positions. The use of the pre-occurrence frontier also covers a step at which the frontier jumps.

For a prime block with introduction index $m = N_p$, all three smoothed values equal $2p = E(m - 1)$. For $n = m, m + 1, m + 2$, the index $m - 1$ is one of $n - 1, n - 2, n - 3$. Equation (20) is unchanged on the $L(n)$ scale by any of these bounded shifts, since $L(n - j)/L(n) \to 1$ and $j = o(L(n))$. The formula follows at the prime and tick positions too.

For completeness, the uniform little-o conclusions here give exactly the quantifiers in (3). Given $\varepsilon > 0$, apply the two excursion propositions with $\delta = \varepsilon/8$. Eventually $L(E(n-1)) \le 2L(n)$, so their contribution is at most $\varepsilon L(n)/4$. Take a common cutoff for the clock estimates shifted by $1, 2, 3$, each with error at most $\varepsilon L(n)/2$, and for the finite seed exclusions. Their sum is at most $\varepsilon L(n)$ at every ordinary index; the shifted clock alone covers the prime blocks. Increasing the common cutoff to at least 3 proves the theorem. □

H Consequences: the diagonal, three lines and short cycles

Theorem 1 has the following consequences for questions recorded in the OEIS [11]. On A064413, Thomas Ordowski conjectured on January 23, 2009, "If $a(n) \neq p$, then almost everywhere $a(n) > n$" and "$\lim \#(a_n > n)/n = 1$". Artur Jasinski, in December 2008, called $a(k) = k$, $2a(k) = 3k$ and $2a(k) = k$ the three critical lines of the sequence. His entry A152458, the indices with $a(k) = k$, records the conjecture that it is finite, and on A152519 (the indices with $2a(k) = 3k$) he conjectured that the number of points on the three lines is finite, asking also whether the listed points are complete. We prove the finiteness, not the completeness. Below, p denotes a prime.

Only one feature of Theorem 1 is new for these purposes: $b(n) > n$ for all large n, the sign of the second-order term. The ratio $b(n)/n \to 1$ was already known [8, 7, Theorem 4.17 and Corollary 12(c), respectively]; it does not decide on which side of the diagonal $a(n)$ lies.

Lemma 8. *For every $\eta > 0$ there is $N_0 \geq 3$ such that $n < b(n) \leq (1+\eta)n$ for all $n \geq N_0$. For every n, exactly one of the following holds: $a(n)$ is an odd prime and $b(n) = 2a(n)$; $a(n) = 3p$ for an odd prime p and $2a(n) = 3b(n)$; or $b(n) = a(n)$.*

Proof. Apply (3) with $\varepsilon = 1/6$: for $n \geq N$, $b(n) \geq n + n/(6\log n) > n$ and $b(n) \leq n + n/(2\log n)$, which is at most $(1+\eta)n$ once $\log n \geq 1/(2\eta)$. The second statement is the definition (2). □

Corollary 9 (The diagonal)**.** *For all sufficiently large n, $a(n) < n$ if $a(n)$ is prime and $a(n) > n$ if $a(n)$ is not prime. Thus, apart from finitely many n, $a(n) > n$ exactly when $a(n)$ is composite, and $a(n) \neq n$. Moreover, as $N \to \infty$,*

$$\#\{k \leq N : a(k) < k\} \sim \frac{N}{2\log N}, \qquad \#\{k \leq N : a(k) > k\} = N - \Big(\frac{1}{2} + o(1)\Big)\frac{N}{\log N}.$$

In particular $\#\{k \leq N : a(k) > k\}/N \to 1$.

Proof. Take N_0 from Lemma 8 with $\eta = 1/3$, and let $n \geq N_0$. If $a(n)$ is prime, it is not 2, since $b(n) > n \geq 3$; so $a(n) = b(n)/2 \leq 2n/3 < n$. If $a(n)$ is not prime, it is not 1 (else $b(n) = 1 < n$), and either $a(n) = b(n) > n$ or $a(n) = 3b(n)/2 > n$.

For the counts, fix $\eta \in (0, 1/3]$ and its N_0. For $N \geq N_0$, the sets $\{k \leq N : a(k) < k\}$ and $\{k \leq N : a(k) \text{ prime}\}$ agree on $[N_0, N]$, and $a(k) = k$ only for $k < N_0$. Let $P(N)$ be the number of $k \in [N_0, N]$ with $a(k)$ prime. Each such k has $a(k) = b(k)/2 \leq (1+\eta)N/2$, and a is injective, so $P(N) \leq \pi((1+\eta)N/2)$. Conversely, let $M = \max\{a(k) : k < N_0\}$. Since a is a permutation, each prime p with $M < p \leq N/2$ is $a(k)$ for one k, and $k \geq N_0$ because $p > M$. Then $k < b(k) = 2p \leq N$, so $P(N) \geq \pi(N/2) - M$. By the prime number theorem $\pi(cN) \sim cN/\log N$ for each fixed $c > 0$, hence

$$1 - o(1) \leq \frac{P(N)}{N/(2\log N)} \leq 1 + \eta + o(1),$$

and η is arbitrary. The second count is N minus the first and minus the finitely many k with $a(k) = k$. □

Ordowski's first conjecture asks only that $a(n) > n$ "almost everywhere" when $a(n)$ is not prime. The first statement, for all but finitely many n, is our stronger form; it implies his conjecture in the density-one reading of "almost everywhere". The last statement is his second conjecture. The count of prime values alone, $\#\{k \leq N : a(k) \text{ prime}\} \sim N/(2\log N)$, already follows from the ratio

form; the new content is that these are, with finitely many exceptions, exactly the indices below the diagonal.

Two related OEIS entries are also finite sets: Ordowski's "exceptional numbers" A155549, the values $a(n)$ that are not prime and satisfy $a(n) \le n-1$ (its terms $16, 26, 49, \ldots$ occur at the indices $n = 17, 27, 50, \ldots$), and A064420, the k with $a(k+1) = k$. Both are listed as "probably complete". By Corollary 9 the indices n with $a(n)$ not prime and $a(n) \le n-1$ form a finite set, so its image A155549 is finite. For the second, $a(n) = n-1$ fails for large n: a prime value has $a(n) \le 2n/3$ and any other value has $a(n) > n$. De Vlieger [4] remarked that it is "not surprising" that A064420 is finite. As below, completeness is not shown.

Corollary 10 (Three lines)**.** *Each of the sets* $\{k : 2a(k) = k\}$, $\{k : a(k) = k\}$ *and* $\{k : 2a(k) = 3k\}$ *is finite.*

Proof. Take N_0 from Lemma 8 with $\eta = 1/3$, and let $n \ge N_0$, so $n < b(n) \le 4n/3$. If $a(n)$ is an odd prime, then $2a(n) = b(n)$, so $n < 2a(n) \le 4n/3$. If $a(n) = 3p$ for an odd prime p, then $2a(n) = 3b(n) > 3n$. Otherwise $a(n) = b(n)$ and $2n < 2a(n) \le 8n/3$. In no case is $2a(n)$ equal to n, $2n$ or $3n$, so all three sets lie in $[1, N_0)$. □

In fact the ratios $a(n)/n$ eventually lie in $(1/2, 2/3] \cup (1, 4/3] \cup (3/2, 2]$, avoiding $1/2$, 1 and $3/2$. No explicit N_0 is extracted, so this does not show that the terms listed in A152458 and A152519 are complete; Amiram Eldar recorded in 2024 that neither sequence has a further term below 10^5 [11].

Lagarias, Rains and Sloane [9, Section 7] note that the permutation appears to have infinitely many finite cycles and infinitely many infinite cycles, and expect both to be hard to prove. The following does not decide either question, but it shows that infinitely many finite cycles, if they exist, have unbounded lengths.

Corollary 11 (Short cycles)**.** *For each* $L \ge 1$*, the permutation* $n \mapsto a(n)$ *of the positive integers has only finitely many cycles of length at most* L*.*

Proof. By Lemma 8, for each n,

$$2^{\sigma(n)} a(n) = 3^{\tau(n)} b(n), \qquad (\sigma(n), \tau(n)) \in \{(0,0), (1,0), (1,1)\}.$$

Choose $\eta > 0$ with $(1+\eta)^L < 2^L/(2^L - 1)$, and N_0 for this η. Suppose a cycle $x_1, \ldots, x_\ell$, with $a(x_i) = x_{i+1}$ for $i < \ell$, $a(x_\ell) = x_1$ and $\ell \le L$, has every $x_i \ge N_0$. Since a permutes the x_i, $\prod a(x_i) = \prod x_i$; with $u = \sum \sigma(x_i)$ and $v = \sum \tau(x_i)$, so $v \le u \le \ell$,

$$2^u \prod_{i=1}^{\ell} x_i = 3^v \prod_{i=1}^{\ell} b(x_i).$$

As $b(x_i) > x_i$, this forces $3^v < 2^u$, so $u \ge 1$ and $3^v \le 2^u - 1$. As $b(x_i) \le (1+\eta) x_i$, the right side is then at most $(2^u - 1)(1+\eta)^\ell \prod x_i$, which is less than $2^u \prod x_i$ because $(1+\eta)^\ell < 2^L/(2^L - 1) \le 2^u/(2^u - 1)$. This contradiction shows that every cycle of length at most L contains an integer below N_0. Distinct cycles are disjoint, so there are fewer than N_0 of them. □

I Numerical illustration

Table 1 gives the normalized error $R(n) = (b(n) - n)\log n/n$, whose limit is $1/3$ by Theorem 1. These values illustrate the formula; they are not used in its proof and do not imply a monotone approach to the limit.

n	$b(n)$	$R(n)$	$1/3$
1,000	1,043	0.297033477	0.333333333
10,000	10,361	0.332493287	0.333333333
100,000	103,088	0.355519138	0.333333333
1,000,000	1,024,686	0.341049694	0.333333333
10,000,000	10,210,804	0.339775904	0.333333333
100,000,000	101,839,957	0.338932605	0.333333333
500,000,000	508,457,544	0.338811220	0.333333333

Table 1: The smoothed EKG error, with absolute rounding error at most $5 \cdot 10^{-10}$ in each displayed ratio.

The included `data/ekg-table.cpp` generates exact integer terms using a bitmap and one least-unused-multiple cursor per prime, as in the method of LRS Section 3. Every eligible value is a multiple of a prime divisor of the preceding term, so the minimum of these cursors gives the next value. The storage bound is checked at every selection; exhausting it aborts rather than truncating the candidate search. A separate direct gcd scan validates the first 4096 terms and rejects an altered prefix. All first 4096 smoothed values are also checked by trial division. The table file records 18 selected indices through $5 \cdot 10^8$, with integer digests in `data/ekg-table-summary.json`.

The ratios were certified by bounding $\log n$ with exact rational sums for $2\operatorname{atanh} z$. After reducing the argument to $[1, 2]$, one uses 60 positive terms with $0 \le z \le 1/3$ and tail at most $2z^{121}/(121(1 - z^2))$. This certifies an absolute error at most 10^{-17} in the recorded ratios and correct rounding to nine decimal places in Table 1.

References

[1] A. C. Cojocaru and M. R. Murty, *An Introduction to Sieve Methods and Their Applications*, LMS Student Texts **66**, Cambridge University Press, 2005, Sections 5.1–5.2. doi:10.1017/CBO9780511615993.

[2] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction—CADE 28*, LNCS **12699**, Springer, 2021, pp. 625–635. doi:10.1007/978-3-030-79876-5_37.

[3] Ch.-J. de la Vallée Poussin, Recherches analytiques sur la théorie des nombres premiers. Première partie: Les nombres premiers en général, *Ann. Soc. Sci. Bruxelles* **20**, deuxième partie (1896), Section 56 (*Digitized collected offprint*, 1897, pp. 67–69). Original offprint.

[4] M. De Vlieger, Constitutive analysis of the EKG sequence, 2021, `https://vincico.com/seq/a064413.html`.

[5] P. Graham, Sharp linear bounds for the EKG sequence, version 1, October 22, 2026, manuscript and Lean sources. doi:10.5281/zenodo.23562177.

[6] J. Hadamard, Sur la distribution des zéros de la fonction $\zeta(s)$ et ses conséquences arithmétiques, *Bull. Soc. Math. France* **24** (1896), 199–220. doi:10.24033/bsmf.545.

[7] T. Haver, Explicit rates for the EKG sequence, version 3, October 4, 2026, manuscript and Lean sources. doi:10.5281/zenodo.23129632.

[8] P. Hofman and M. Pilipczuk, A few new facts about the EKG sequence, *J. Integer Sequences* **11** (2008), Article 08.4.2.

[9] J. C. Lagarias, E. M. Rains, and N. J. A. Sloane, The EKG sequence, *Experimental Math.* **11** (2002), 437–446. doi:10.1080/10586458.2002.10504486; arXiv:math/0204011v1.

[10] The mathlib Community, The Lean mathematical library, in *Proceedings of CPP '20*, ACM, 2020, pp. 367–381. doi:10.1145/3372885.3373824; arXiv:1910.09336.

[11] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entries A064413, A064420, A152458, A152519, A155549 and A256417.

[12] A. Kontorovich, T. Tao, et al., *PrimeNumberTheoremAnd* (Prime Number Theorem and More), Lean sources, revision `c39a751132c88b6e8080b74c74023fd95b3d8be0`, 2026. https://github.com/AlexKontorovich/PrimeNumberTheoremAnd.

2020 *Mathematics Subject Classification*: Primary 11B83; Secondary 11A05, 68V15.
Keywords: EKG sequence, greedy permutation, prime number theorem, asymptotic formula, formal verification.
Concerned with sequences A064413, A152458, A152519 and A256417.